

Data-driven collaborative optimization of three-dimensional highway alignments for life-cycle cost and mixed fuel–electric traffic energy

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Abstract

Highway alignment design couples plan geometry, vertical profile, terrain, structures, urban development and long-term vehicle operation. This study develops a data-driven bi-objective framework that makes these interactions spatially explicit. A quasi-natural digital elevation model (DEM), OpenStreetMap (OSM) transport, water and building layers, and 14,018 GNSS points are fused in one corridor model. Fifty low-frequency plan modes and 224 active grade variables are optimized collaboratively. Life-cycle cost covers land, basic works, structures, earthwork and maintenance; the second objective monetizes 30-year fuel and electricity consumption for a mixed fleet. Candidate–OSM intersections trigger crossing bridges, a connected mountain zone triggers ecological tunnelling, and dense built-up cores are excluded. An improved Jellyfish Search supports weighted-sum front approximation, followed by feasibility, budget and non-dominance gates before entropy-based compromise selection.

The model is applied to 22.46 km of Guangzhou North Ring Expressway. With endpoint elevations anchored, spectral search width $W = 500$ m and density constraints active, the selected alignment reduces life-cycle cost from 2.6406 to 2.4208 billion RMB (8.33%) and monetized traffic energy from 1.3946 to 1.3528 billion RMB (3.00%). It retains a 421-m minimum radius, zero hard-constraint penalty and zero Tier-2 exposure; Tier-1 exposure decreases from 2.010 to 1.591 km. The route is 12 m longer and uses more earthwork, while structure cost falls by 237 million RMB, revealing a traceable physical trade-off rather than a length artefact. Earlier free-endpoint, unequal-budget experiments are exploratory only. Current-model multi-seed and equal-NFE replication is required before design-grade use.

Keywords: highway alignment optimization, digital elevation model, OpenStreetMap, improved Jellyfish Search, life-cycle cost, mixed traffic energy, horizontal-vertical coordination

1. Introduction

Highway alignment is an early design decision with long-lived consequences. A small lateral shift changes the terrain crossed, land take, bridge and tunnel demand, earthwork balance, maintenance exposure and the grades experienced by vehicles. These effects are coupled: shortening a horizontal path may worsen the vertical profile; avoiding a mountain may enter a dense urban zone; lowering earthwork can require a bridge; and a profile that appears energy-efficient in one direction can become infeasible at its network tie-ins. The practical problem is therefore not a curve-fitting exercise but a constrained, spatially explicit, multi-objective design problem.

Classical formulations established the value of simultaneous horizontal and vertical optimization [1, 2]. Subsequent research introduced corridor models, mixed-integer formulations, environmental objectives and preference-aware multi-objective search [3, 4, 5, 6, 7]. Recent work has coupled alignment generation with BIM, GIS, deep reinforcement learning, sampling-based planning, and hybrid reinforcement-learning/metaheuristic search [8, 9, 10, 11, 12]. A comprehensive review by Song et al. [13] nevertheless identifies enduring gaps between algorithmic models and the heterogeneous spatial constraints encountered in engineering design.

Three gaps motivate this study. First, many models use a single centerline terrain profile or stylized land-cost grid. This prevents horizontal displacement from changing the terrain and can erase the main economic reason for joint optimization. Second, bridges, tunnels and urban avoidance are commonly imposed as fixed lengths or discontinuous penalties. Such simplifications allow an optimizer to exploit bookkeeping rules rather than discover an engineering alternative. Third, large nonlinear decision spaces remain difficult for population-based solvers. A solver may report a low objective while retaining localized curvature or grade-change violations that disappear in an averaged penalty.

This paper extends the mathematical base in Lin [14] while replacing its NSGA-II-centered and centerline-based implementation with a current, data-driven experiment. The main contributions are:

- 33 (1) a spatially explicit digital corridor that fuses GNSS, quasi-natural DEM,
34 OSM transport/water features and OSM-derived building-density tiers;
- 35 (2) an engineering-oriented life-cycle model in which crossing bridges, ecolog-
36 ical tunnelling, earthwork-to-structure substitution and endpoint tie-ins
37 are endogenous to the three-dimensional geometry;
- 38 (3) a nominally 275-slot, 274-active-variable collaborative parameterization
39 with constructive endpoint and grade handling, coupled to a bi-objective
40 cost–energy decision process;
- 41 (4) an IJS solver that combines corrected Tent initialization, domain-scaled
42 Lévy exploration and differential-evolution refinement; and
- 43 (5) an auditable evidence chain that separates the current engineering result
44 from exploratory experiments conducted during earlier model develop-
45 ment.

46 Figure 1 summarizes the proposed pipeline. The visual distinction be-
47 tween data, digital corridor, coupled model, optimizer and validated align-
48 ment is intentional: each block has an auditable counterpart in the experi-
49 ment repository.

50 2. Related research

51 2.1. Three-dimensional alignment optimization

52 Early simultaneous models demonstrated that horizontal and vertical de-
53 cisions should not be optimized independently [1]. Evolutionary encoding
54 subsequently made non-convex three-dimensional search practical [2]. GIS-
55 based studies expanded the feasible region and allowed environmental layers
56 to influence routing [3, 15]. Corridor-constrained horizontal search [4] and
57 rigorous vertical-alignment formulations [7] improved mathematical tractabil-
58 ity, while bi-objective road and railway models incorporated cost, hazard and
59 ecological impacts [5, 16, 17].

60 Recent end-to-end models integrate BIM or intelligent path planners,
61 while newer hybrid schemes combine policy optimization with population-
62 based search [8, 11, 12]. These methods improve automation but do not
63 eliminate the need for transparent engineering accounting. In particular, a
64 spatial intersection must lead to a structure or clearance decision, terrain

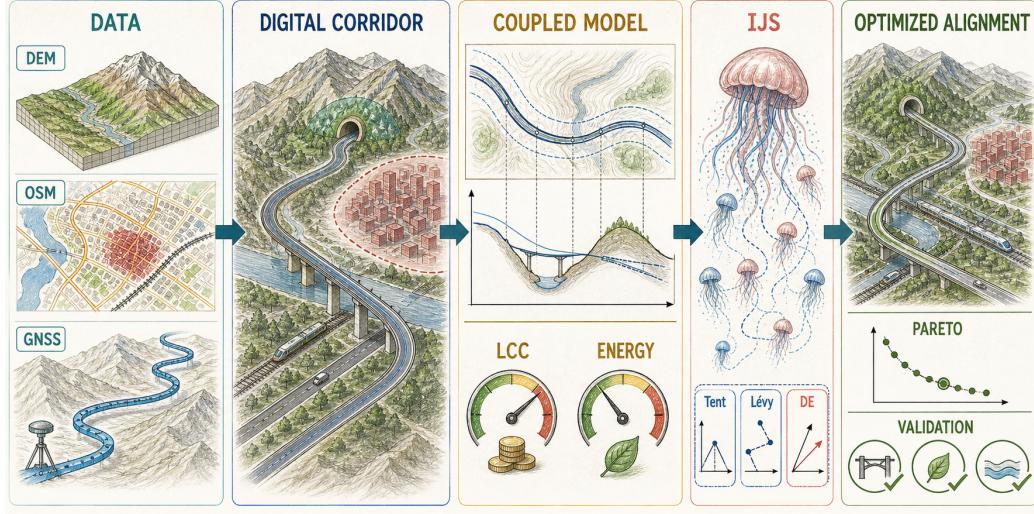


Figure 1: Data-driven workflow for collaborative three-dimensional highway alignment optimization. The hand-drawn scientific illustration highlights the information flow; the mathematical definitions are given in Sections 3 and 4.

65 sampling must follow the candidate line, and existing network connections
66 must be preserved. This paper therefore prioritizes traceable geometry-to-
67 cost mappings over an opaque surrogate objective.

68 2.2. Life-cycle cost and traffic energy

69 The underlying life-cycle model follows the categories in Lin [14]: land
70 acquisition, bridges/tunnels, road works, maintenance and earthwork. Vehi-
71 cle operation is relevant because geometric consistency affects emissions and
72 energy demand [18]. Electric-vehicle energy is especially sensitive to grade,
73 mass, auxiliaries and regenerative behavior [19]. Rather than adding fuel and
74 electricity in incompatible physical units, the present framework monetizes
75 both over the analysis period. This does not claim that money is a complete
76 environmental metric; it supplies a consistent decision scale while physical
77 fuel and electricity components remain reportable.

78 2.3. Optimization algorithms and objective decision

79 NSGA-II remains a standard multi-objective benchmark [20], but direct
80 evolutionary search becomes expensive as the number of profile variables
81 increases. Jellyfish Search models ocean-current and active/passive motion
82 [21]. The present IJS adds three complementary mechanisms: a Tent map

for initial coverage, Lévy flights for non-local exploration and differential evolution (DE) for vector-wise refinement [22]. The mechanisms are not assumed beneficial a priori; Section 5.3 measures their isolated and combined contributions.

Entropy weighting is based on the information concept of Shannon [23]. Here it is applied only after feasibility and non-dominance screening. Thus a high weight cannot compensate for a geometric violation, and an infeasible solution cannot be selected merely because its cost is low.

3. Data-driven mathematical formulation

3.1. Study corridor and data fusion

The case is the Xunfengzhou–Huangcun portion of Guangzhou North Ring Expressway, a two-way six-lane urban expressway with a measured centerline length of 22.462 km. The input comprises 14,018 GNSS points with plan coordinates and elevations. These points define the existing scheme M-A, the fixed plan endpoints and the vertical tie-in elevations.

Terrain is sampled from AWS Terrain Tiles [24]. Sixty zoom-level-14 tiles are decoded at approximately 8.8 m pixel spacing. Because the available surface represents the built road, a ± 60 -m road mask is removed and interpolated from neighboring terrain to obtain a quasi-natural ground. Its correlation with the measured centerline elevation is 0.915 after a constant vertical-datum correction. This reconstruction is imperfect but applies the same terrain accounting to the existing and optimized alternatives.

OSM [25, 26] supplies roads, railways, watercourses and building footprints. After removing the North Ring Expressway and parallel facilities, the obstacle set contains 4,807 road features, 857 railway features and 338 river/canal features. Building footprints are rasterized on a 25-m grid and smoothed with a 200-m kernel. Thresholds are calibrated against the maximum density already traversed by M-A: Tier 2 is a hard exclusion, Tier 1 is passable with a soft discouragement, and Tier 0 is unrestricted. Figure 2 shows the aligned layers.

3.2. Collaborative decision variables

Let the measured plan line be $\mathbf{r}_0(s) = [x_0(s), y_0(s)]^\top$ with unit normal $\mathbf{n}(s)$, $s \in [0, L_0]$. A candidate plan is represented by a normal offset

$$\delta(s) = \sum_{k=1}^{50} a_k \sin\left(\frac{k\pi s}{L_0}\right), \quad |a_k| \leq \frac{W}{k^2}, \quad (1)$$

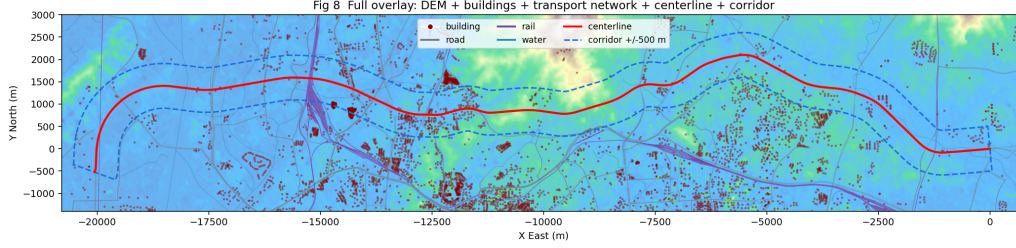


Figure 2: Case-study data layers in the local engineering coordinate system: DEM, OSM buildings, roads, railways, watercourses, measured centerline and a displayed ± 500 -m context band. In the optimizer W is the first sine-mode amplitude, not a pointwise hard offset bound.

116 where $W = 500$ m in the main experiment and $\mathbf{r}(s) = \mathbf{r}_0(s) + \delta(s)\mathbf{n}(s)$.
 117 The sine basis fixes both endpoints and attenuates high-frequency curvature.
 118 It implies the conservative spectral envelope $|\delta| \leq W \sum_{k=1}^{50} k^{-2} \approx 1.625W$;
 119 hence W is not a pointwise corridor boundary. The selected M-C has a
 120 maximum sampled offset of approximately 184 m. Offset control points are
 121 reconstructed by a cubic spline and evaluated on a 10-m grid. For recon-
 122 structed coordinates $(x(s), y(s))$,

$$\kappa(s) = \frac{|x'(s)y''(s) - y'(s)x''(s)|}{[x'(s)^2 + y'(s)^2]^{3/2}}, \quad R(s) = \kappa(s)^{-1}. \quad (2)$$

The vertical profile uses $M = 225$ stations (approximately 100 m apart). Normalized variables generate segment grades, but the start and end design elevations z_0 and z_L are tied to the measured road. For raw grade perturbations $h_i \in [-1, 1]$, segment lengths Δs_i and maximum grade $g_{\max} = 0.04$, define

$$\bar{h} = \frac{\sum_i h_i \Delta s_i}{\sum_i \Delta s_i}, \quad g_{\text{req}} = \frac{z_L - z_0}{\sum_i \Delta s_i}, \quad (3)$$

$$g_i = g_{\text{req}} + \lambda g_{\max} (h_i - \bar{h}), \quad (4)$$

where $\lambda \in [0, 1]$ is the largest scaling that retains $|g_i| \leq g_{\max}$. Consequently $\sum_i g_i \Delta s_i = z_L - z_0$ exactly. The implementation retains a legacy start-elevation gene in its 225-slot profile block, but ignores it when z_0 is tied. The stored vector therefore has 275 slots but 274 active variables (50 modes and 224 grades); the roughly 150 reconstruction controls are not optimization variables.

3.3. Life-cycle cost

The first objective is

$$\min C = C_R + C_B + C_S + C_Q + C_{TU}, \quad (5)$$

where C_R is land acquisition, C_B bridge/tunnel cost, C_S basic subgrade and pavement cost, C_Q discounted maintenance and C_{TU} earthwork/haulage. With road width B , length L , one mu = 666.67 m², bridge/tunnel lengths L_B, L_T and invariant ramp cost C_{ramp} , the implemented accounting is

$$C_R = c_{land} \frac{LB}{666.67} + 5000 \frac{L}{1000}, \quad (6)$$

$$C_B = c_B L_B + C_{ramp} + c_T L_T, \quad C_S = (c_{sg} + c_{pv})L, \quad (7)$$

$$C_Q = \sum_{y=1}^T \frac{1}{(1+r)^y} \sum_i \Delta s_i [\gamma + 365 Q_y \tau + c_{soil} \Phi_i]. \quad (8)$$

Here Φ_i is the weighted cut/fill slope-exposure term in the implementation; it is zero on structure-exempt segments. Unit rates follow the source thesis and the 2025 Guangzhou governmental estimation guideline [27]. The analysis period is 30 years with a 5% discount rate.

At station j , let $h_j = z_j - z_j^g$ be the design-to-ground height, B the formation width and m the side-slope ratio. The approximate cross-sectional earthwork area is

$$A_j = B|h_j| + mh_j^2. \quad (9)$$

Volumes are integrated by the average-end-area method,

$$V^\pm = \sum_j \frac{A_j^\pm + A_{j+1}^\pm}{2} \Delta s_j, \quad C_{TU} = E_L + \sum_j \Delta s_j \min(c_{ew} A_j, c_{str}), \quad (10)$$

with cut/fill unit rates selected by the sign of h_j . If fill/cut cost exceeds the corresponding bridge/tunnel cost, or if $|h_j| > 30$ m, structural substitution

Table 1: Principal current-model parameters used in the case study.

Group	Parameter	Value
Geometry	speed / width / R_{min} / g_{max}	100 km/h / 25.5 m / 400 m / 4%
Life cycle	T / discount rate	30 yr / 5%
Land/basic works	c_{land} / c_{sg} / c_{pw}	58,641 RMB/mu / 200 / 30,000 RMB/m
Structures	c_B / c_T / E_{ext}	156 / 270 million RMB/km / 75 m
Earthwork	cut / fill / side slope	30 / 25 RMB/m ³ / 1:1.5
Traffic	AADT / EV share	30,000 veh/day / 0.30
Energy	fuel / electricity price	8 RMB/L / 0.8 RMB/kWh
Search	population / iterations / weights	200 / 1,000 / 21

141 is triggered. The selected structure cost replaces rather than supplements
 142 earthwork in that interval, preventing double counting and “free floating
 143 road” solutions.

144 For each candidate plan, intersections with OSM road classes primary
 145 and above, railways and watercourses trigger bridges. If θ_q is the crossing
 146 angle and b_q the nominal obstacle width, the required crossing interval is

$$\ell_q = \frac{b_q}{\max(\sin \theta_q, \sin 15^\circ)} + 2E_{\text{ext}}, \quad (11)$$

147 where $E_{\text{ext}} = 75$ m is the approach extension. Gaps under 100 m are merged.
 148 Ramp-function cost is calibrated from the seven existing interchanges and
 149 kept invariant, whereas mainline bridge length is geometric. A connected
 150 terrain region above 70 m, closed morphologically over 500 m, defines the
 151 Baiyun ecological mandatory-tunnel zone. Its area (19.2 km²) and M-A
 152 crossing length (1.28 km) agree with independent project information (about
 153 21 km² and 1.35 km, respectively).

154 3.4. Mixed fuel–electric traffic energy

155 The second objective monetizes lifetime operation of a mixed fleet:

$$\min E = \sum_{y=1}^{30} \frac{365 Q_y}{(1+r)^y} [(1 - p_{EV})c_f e_f(\mathbf{x}) + p_{EV}c_e e_e(\mathbf{x})], \quad (12)$$

156 where Q_y is AADT, $p_{EV} = 0.30$, c_f and c_e are fuel and electricity prices, and
 157 e_f, e_e are per-vehicle route consumption. The baseline uses AADT=30,000
 158 vehicles/day, gasoline/electricity prices of 8 RMB/L and 0.8 RMB/kWh, and
 159 the same discount rate as cost.

For a fuel vehicle, the traction load includes aerodynamic, rolling, grade and residual lateral forces. With speed v , mass m_v , aerodynamic coefficient C_d , frontal area A_f , grade angle α and curve radius R , these are

$$F_{air} = \frac{1}{2}\rho C_d A_f v^2, \quad F_r = C_r m_v g \cos \alpha, \quad (13)$$

$$F_g = m_v g \sin \alpha, \quad F_a = \max \left(0, \frac{m_v v^2 / R - m_v g S_p}{N_v C_s} \right). \quad (14)$$

For fuel vehicles the implementation converts external load to horsepower and fuel volume as

$$P_{ex,i} = \max \left(0, \frac{(F_{air} + F_r + F_g + F_a)v}{736\eta_f} \right), \quad (15)$$

$$e_f(\mathbf{x}) = \sum_i \phi(HP_{in} + P_{ex,i}) \frac{\Delta s_i}{v} / 1000, \quad (16)$$

160 where e_f is L/vehicle-trip, $\phi = 0.03$ ml/(s hp), $HP_{in} = 27$ hp and $\eta_f =$
 161 0.85 . The printed 10^{-3} multiplier in the source expression for F_a is omitted:
 162 retaining it makes lateral resistance roughly five orders too small and removes
 163 the plan-curvature energy channel. This dimensional correction is explicitly
 164 disclosed rather than silently absorbed into calibration.

165 Electric consumption uses signed segment work $W_i = (F_{air} + F_r + F_g +$
 166 $F_a)\Delta s_i$:

$$e_e(\mathbf{x}) = \frac{1}{3.6 \times 10^6} \sum_i \begin{cases} W_i/\eta_e, & W_i \geq 0, \\ W_i\eta_e, & W_i < 0, \end{cases} + 0.15L_{km}, \quad \eta_e = 0.90 \times 0.95. \quad (17)$$

167 Fuel and electric use are retained separately before monetary aggregation.
 168 The current main experiment evaluates a representative travel direction; fix-
 169 ing both endpoint elevations removes the former net-downhill bookkeeping
 170 exploit. A bidirectional mean is identified as a required extension in Sec-
 171 tion 6.4.

172 3.5. Constraints and decision rule

173 The hard engineering set includes fixed plan/profile endpoints, $R(s) \geq$
 174 400 m for 100 km/h design speed, $|g_i| \leq 4\%$, $|g_{i+1} - g_i| \leq 3 \times 10^{-4}\Delta s_{ctrl}$,
 175 crossing skew angle $\theta \geq 15^\circ$, and zero Tier-2 penetration. Values follow JTG
 176 D20-2017 [28] where directly applicable. W controls sine-mode amplitudes

177 but is not a pointwise corridor constraint; the selected current line remains
 178 inside the displayed ± 500 -m context band. Local violations use the sum
 179 of mean and maximum relative violation, so a short severe defect cannot
 180 disappear in a long route average.

181 For any relative-violation vector $\mathbf{v}$, define $H(\mathbf{v}) = \text{mean}(\mathbf{v}) + \max(\mathbf{v})$.
 182 The implemented penalty is

$$P = \rho_t \{30H(\mathbf{v}_R) + 10H(\mathbf{v}_{\Delta g}) + 10H(\mathbf{v}_{skew}) + 10H(\mathbf{v}_{D2})\}, \quad (18)$$

183 where $v_R = [400 - R]_+/400$, $v_{\Delta g}$ and v_{skew} are normalized excesses, and
 184 $v_{D2} = \min(d_{D2}/100, 5)$. The schedule is $\rho_t = 0.3$ in the first half and 3.0
 185 thereafter. Tier 1 is deliberately excluded from P and appears only as the
 186 soft term in Eq. (19).

187 For objectives C and E , a common baseline population supplies reference
 188 scales C_{ref} and E_{ref} . Entropy weights w_C, w_E produce

$$F(\mathbf{x}) = w_C \frac{C(\mathbf{x})}{C_{ref}} + w_E \frac{E(\mathbf{x})}{E_{ref}} + P(\mathbf{x}) + 0.22 \frac{L_{T1}(\mathbf{x})}{L(\mathbf{x})}. \quad (19)$$

189 For an $n \times 2$ objective matrix, min-max normalized entries z_{ij} give $p_{ij} =$
 190 $z_{ij} / \sum_i z_{ij}$, entropy $e_j = -(\ln n)^{-1} \sum_i p_{ij} \ln(p_{ij} + 10^{-12})$, and $w_j = (1 -$
 191 $e_j) / \sum_k (1 - e_k)$. Weights computed from a common initial population define
 192 the search scalarization. They are recomputed on the surviving front for the
 193 final compromise; these are two distinct uses of entropy weighting.

194 IJS solves 21 independent weighted-sum problems, so the resulting set is
 195 a weighted-sum Pareto approximation and can miss unsupported points on
 196 a non-convex front. The final decision is selected by four gates: hard feasi-
 197 bility, project-screening budget $C \leq 1.1C_{M-A}$, non-dominance, and entropy-
 198 weighted compromise. The 10% budget margin is an explicit planning as-
 199 sumption rather than a statutory threshold; its decision sensitivity remains
 200 to be tested.

201 4. Improved Jellyfish Search

202 4.1. Mechanisms

203 The original JS alternates an ocean-current update toward the best so-
 204 lution with active/passive movement within the population [21]. The IJS
 205 implementation adds:

- 206 • **Independent Tent initialization.** Each individual uses an independent
 207 chaotic chain with $\mu = 1.99$. The exact value $\mu = 2$ is avoided because
 208 finite-precision binary arithmetic collapses to zero; a single shared chain
 209 is avoided because it creates correlated individuals.
- 210 • **Domain-scaled Lévy flight.** The heavy-tailed perturbation is scaled
 211 by $0.01(\mathbf{u} - \mathbf{l})$. The former unscaled implementation had only a 0.48%
 212 acceptance rate because a unit step is not meaningful across heterogeneous
 213 variable ranges.
- 214 • **DE refinement.** Mutation and crossover with $CR = 0.5$ form a trial
 215 vector from population differences, followed by elitist acceptance [22].

216 For reproducibility, the time control is $A_r = (1 - t/T)(2u - 1)$. If $|A_r| \geq$
 217 0.5 , the main trial is $\mathbf{x}_i + \mathbf{u} \odot (\mathbf{x}_{best} - 3u\bar{\mathbf{x}})$; otherwise the code applies active
 218 motion toward a fitter random peer or passive motion $\mathbf{x}_i + 0.1\mathbf{u} \odot (\mathbf{u}_b - \mathbf{l}_b)$.
 219 The Lévy trial is

$$\mathbf{x}_i^L = \mathbf{x}_i + \mathbf{u} \odot [0.01(\mathbf{u}_b - \mathbf{l}_b) \odot \mathbf{L}_{1.5}], \quad (20)$$

220 and the DE trial uses $\mathbf{m}_i = \mathbf{x}_i + u(\mathbf{x}_{r1} - \mathbf{x}_{r2})$, binomial crossover with
 221 $CR = 0.5$, and greedy replacement. Out-of-bound coordinates are reflected
 222 across the opposite bound and then clipped. Every main, Lévy and DE trial
 223 is evaluated before elitist acceptance.

224 Each IJS generation therefore has about three population-equivalent eval-
 225 uation phases; this is reported rather than comparing equal generation counts
 226 as equal effort. A two-phase penalty schedule uses 0.3 of the nominal penalty
 227 in the first half and 3.0 in the second half, allowing exploration near feasibility
 228 boundaries before final constraint recovery. Figure 3 gives the full process.

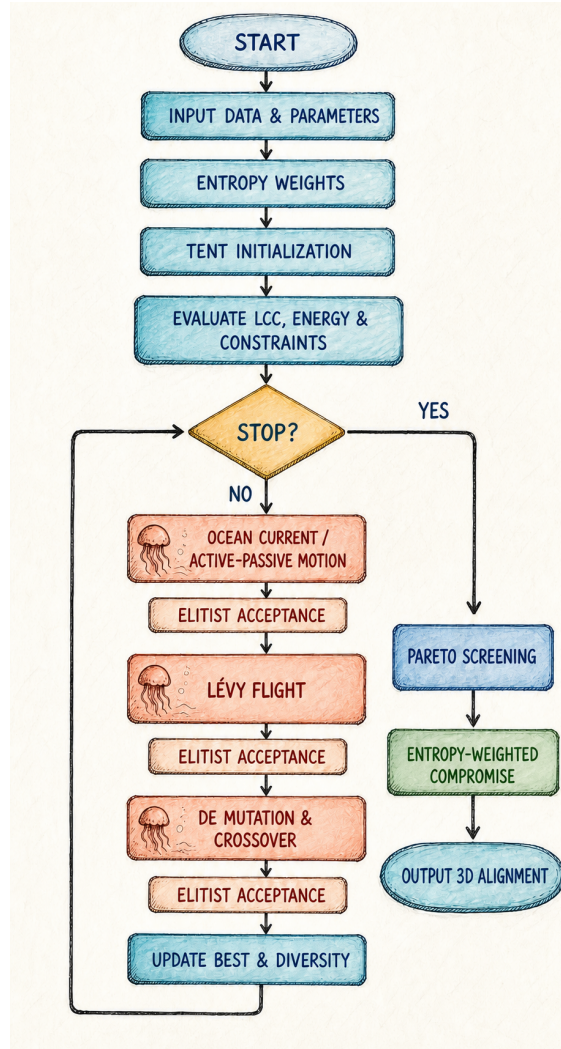


Figure 3: IJS solution and decision flow. Each operator is followed by elitist acceptance; the stopping branch proceeds to Pareto screening and entropy-weighted compromise selection, not geometric knee selection.

4.2. Experimental protocol

The scheme experiment uses population 200, 1,000 IJS iterations and 21 Pareto weights (a later visual sweep uses 40). Objectives are evaluated at approximately 10-m spacing. The baseline, cost-only and joint alternatives use the same data and reference scales.

The historical ablation fixes the measured plan and uses a nominal 225-variable free-elevation profile model. Five variants (JS, +Tent, +Lévy, +DE and complete IJS) have 30 runs, population 200 and 500 iterations. The historical benchmark compares IJS, JS, NSGA-II, real-coded GA, PSO and GWO at six profile spacings (95–500 stored variables), with 10 seeds per pair. Reported p values come from independent-sample Mann–Whitney/rank-sum tests, not paired signed-rank tests. These experiments predate endpoint anchoring and density tiers. Equal generations also imply unequal NFE: approximately 100,200 for JS, 200,200 for one-added-operator variants and 300,400 for full IJS in the 500-generation ablation. They are therefore retained only as transparent development evidence and cannot establish equal-budget superiority of IJS for the current model. The same boundary applies to the 237 historical re-optimization sensitivity points.

5. Results

5.1. Current engineering scheme

Table 2 and Figure 4 report the current main result: $W = 500$ m, density constraints active and endpoint elevations anchored. M-C reduces C by 8.33% and E by 3.00%. The route is 12 m longer (+0.05%), demonstrating that the saving is not an artificial consequence of a shorter path. Bridge/tunnel cost decreases by 237 million RMB, while earthwork rises by 15 million RMB. The net effect is physically interpretable: the optimizer accepts more earthwork and a slightly longer route to reduce structural exposure and operating energy.

The reported diagnostic slope-hazard index is $Q = \sum_{u=1}^5 W_u S_u$ with scores $S_u \in \{1, 3, 5\}$ and weights (0.30, 0.15, 0.20, 0.20, 0.15) for topography, gully density, rainfall, geology and vegetation. The topographic score is the worse of grade-angle and cut/fill-height classes; the other four are corridor-level calibrated constants. Hence Q is a comparative screening diagnostic, not a probabilistic slope-failure model or an optimization objective.

The selected line crosses 1.591 km of Tier 1 and no Tier 2. Figure 5 verifies this against the density field. It also avoids treating the existing route

Table 2: Current endpoint-anchored, density-constrained result.

Metric	M-A existing	M-C optimized	Change
Life-cycle cost (10^8 RMB)	26.4061	24.2078	−8.33%
Traffic energy (10^8 RMB)	13.9460	13.5278	−3.00%
Fuel component	11.1072	10.7911	−2.85%
Electricity component	2.8388	2.7367	−3.60%
Length (km)	22.4618	22.4741	+0.05%
Minimum radius (m)	422	421	feasible
Slope-hazard index	3.309	3.298	−0.35%
Tier-1 exposure (km)	2.010	1.591	−20.85%
Tier-2 exposure (km)	0	0	feasible
Penalty	0	0	feasible

Cost and energy are 30-year discounted monetary quantities. Tier-1 is passable with a soft term; Tier-2 is forbidden.

as an exception: Tier thresholds are calibrated so that a density already traversed by M-A is evidence of passability, while denser connected cores remain forbidden.

The current Pareto set (Figure 6) contains costly structure-heavy energy endpoints as well as the practically relevant cluster. Feasibility and the 10% budget gate remove those extremes before entropy selection. The chosen M-C is therefore a constrained compromise, not simply the minimum of a globally normalized weighted sum.

Figure 7 provides the spatial interpretation. The present M-C contains approximately 4.08 km of OSM-triggered crossing bridges and 0.75 km in the Baiyun ecological tunnel zone. These values differ from earlier repository figures because the current density-constrained solution is used here.

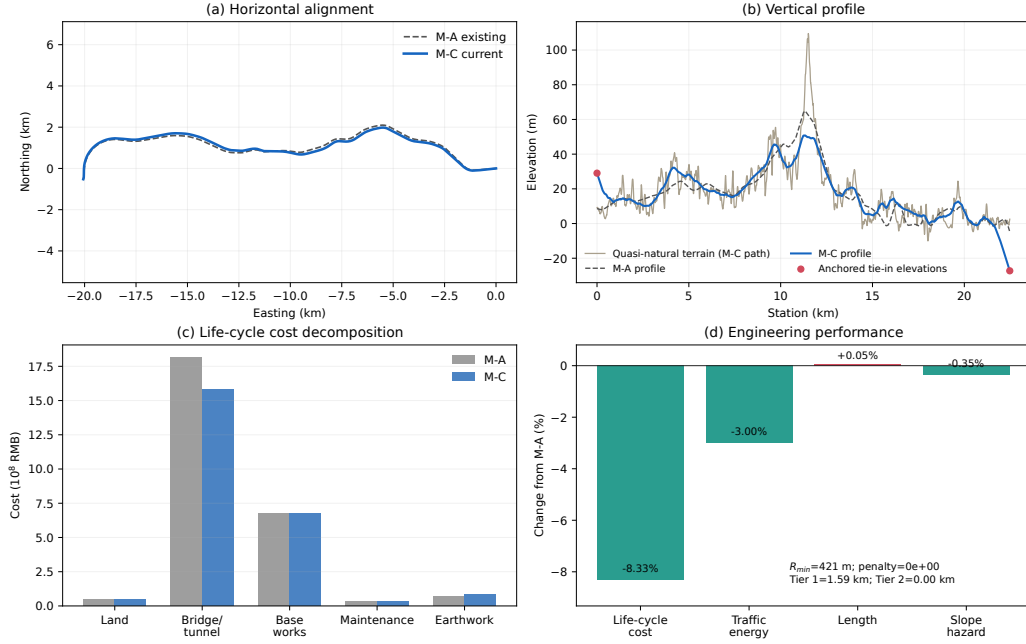


Figure 4: Current M-A/M-C comparison: (a) plan, (b) endpoint-anchored profile, (c) life-cycle cost decomposition and (d) headline changes. The increased earthwork and small length increase are shown explicitly.

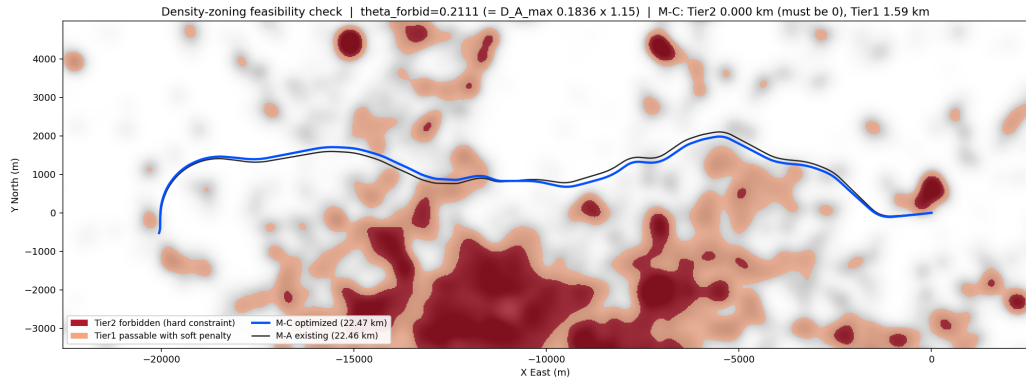


Figure 5: Building-density feasibility check. The current M-C line avoids Tier-2 cores and reduces Tier-1 exposure to 1.59 km.

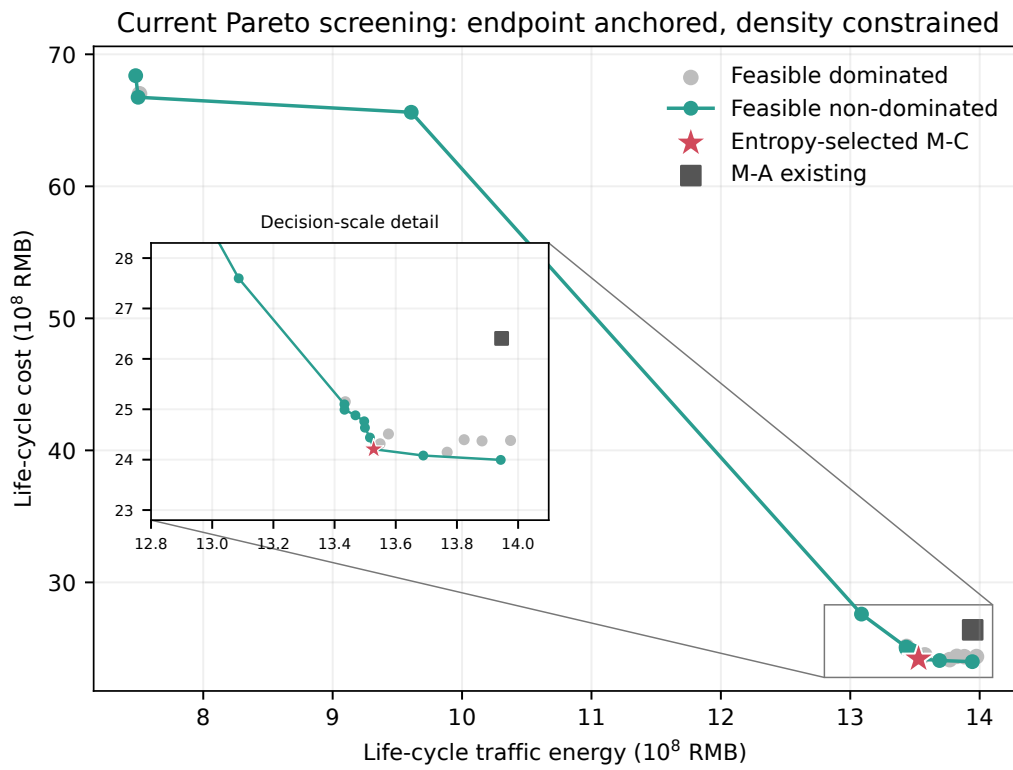


Figure 6: Feasible and non-dominated screening under the current endpoint and density constraints.

Current M-C alignment on quasi-natural DEM (endpoint anchored; density constrained)

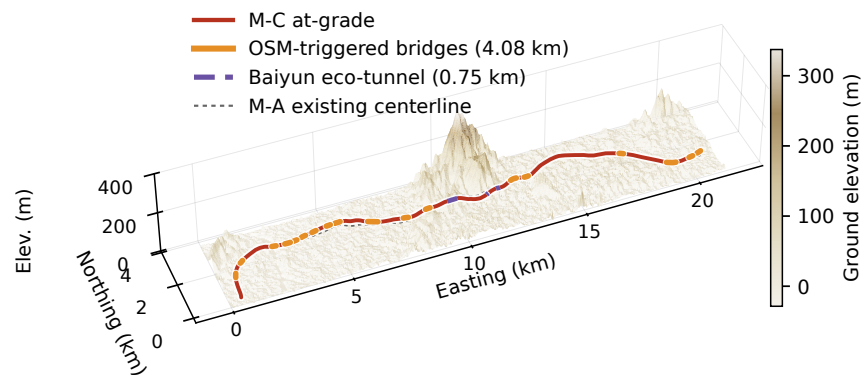


Figure 7: Current M-C on the quasi-natural DEM. At-grade, OSM-triggered bridge and ecological-tunnel intervals are distinguished; M-A is shown as a dashed reference.

277 *5.2. Historical pre-density control: exploratory evidence*

278 A controlled experiment completed before endpoint anchoring and density
279 tiers used identical objective references and equal evaluation budgets for joint
280 and two-stage methods. The existing line had $C = 26.4061$, $E = 13.9460$.
281 The two-stage method obtained $C = 23.1582$ and $E = 13.4569$, apparently
282 better than the joint result ($C = 24.2239$, $E = 13.4766$). However, its mini-
283 mum radius was 397 m and penalty 7.36×10^{-2} , whereas the joint solution
284 had 401 m and zero penalty. The lower two-stage objectives are therefore
285 not admissible evidence of superiority. Freezing the plan after stage one
286 removed degrees of freedom needed near the curvature boundary. This his-
287 torical result motivates collaborative search, but must be rerun with current
288 endpoint/density constraints before it can quantify a current-model effect.

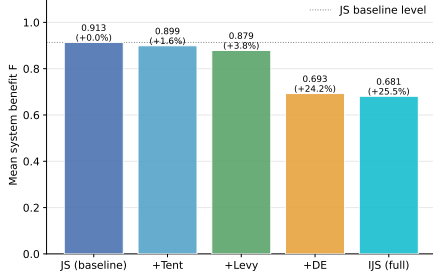
289 *5.3. Historical same-generation IJS ablation*

290 Table 3 shows the 30-run historical ablation. At the same generation
291 count, IJS lowers mean F from 0.9133 to 0.6808 and reaches the final JS level
292 in 187 rather than 500 generations. The observed 25.45% gap is confounded
293 with about three times the NFE and an earlier free-endpoint objective; it
294 is not an equal-budget improvement estimate. DE has the largest observed
295 isolated gap (24.17%) but also doubles NFE. Lévy and Tent show 3.79% and
296 1.58% gaps. The unadjusted rank-sum p values are retained for audit, not
297 confirmatory inference.

Table 3: Historical same-generation ablation. NFE differs by variant; p values are unadjusted independent-sample rank-sum tests.

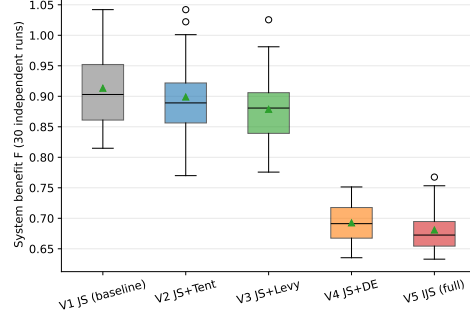
Variant	Mean F	SD	F_{100}	Gain vs JS	p
JS	0.9133	0.0637	2.547	—	—
JS+Tent	0.8989	0.0605	2.227	1.58%	0.464
JS+Lévy	0.8787	0.0558	2.095	3.79%	0.0555
JS+DE	0.6926	0.0315	1.679	24.17%	3.02×10^{-11}
IJS	0.6808	0.0340	1.292	25.45%	3.02×10^{-11}

Fig. A2 Contribution decomposition of the three improvement components



(a) Component contribution.

Fig. A3 Box plots of 30 independent runs for each ablation variant



(b) Run-to-run distribution.

Figure 8: Ablation results. Lower F is better.

298 Mechanism instrumentation clarifies the result. Independent Tent chains
 299 replace 132 of 200 random initial individuals. DE still contributes about 0.018
 300 improvement in the final 100 generations, indicating local refinement rather
 301 than only early acceleration. A pre-defined “escape after 20 stagnant gener-
 302 ations” statistic is not applicable because complete IJS rarely experiences a
 303 20-generation stagnant interval; reporting it as zero would incorrectly imply
 304 failure.

305 5.4. Historical same-generation algorithm benchmark

306 At equal generation counts, IJS has the lowest historical mean F at all
 307 six scales (Table 4). These values describe an earlier free-endpoint model and
 308 IJS uses roughly three population-equivalent evaluations per generation, so
 309 the table cannot establish equal-budget superiority. At PJ6, IJS and JS have
 310 10/10 feasibility whereas the other four algorithms have 0/10; this feasibility

Table 4: Historical objective mean (SD) over ten runs. Equal generations do not mean equal NFE.

Algorithm	PJ1 (95)	PJ2	PJ3	PJ4	PJ5	PJ6 (500)
IJS	.6617 (.0051)	.5986 (.0045)	.7087 (.0045)	.7090 (.0033)	.7778 (.0078)	.8333 (.0019)
JS	.6771 (.0052)	.6104 (.0049)	.7237 (.0063)	.7215 (.0066)	.7954 (.0020)	.8563 (.0060)
NSGA-II	.6792 (.0045)	.6240 (.0057)	.7282 (.0053)	.7287 (.0080)	.8104 (.0084)	12.8859 (.7330)
GA	.7103 (.0088)	.6478 (.0171)	.7515 (.0090)	.7489 (.0071)	5.3906 (.5481)	25.6024 (1.0630)
PSO	.7023 (.0116)	.6370 (.0113)	.7436 (.0070)	.7632 (.0072)	9.1988 (.9109)	32.7437 (.6336)
GWO	.6843 (.0065)	.6194 (.0070)	.7233 (.0046)	.7276 (.0070)	1.1104 (.0158)	3.2157 (1.5706)

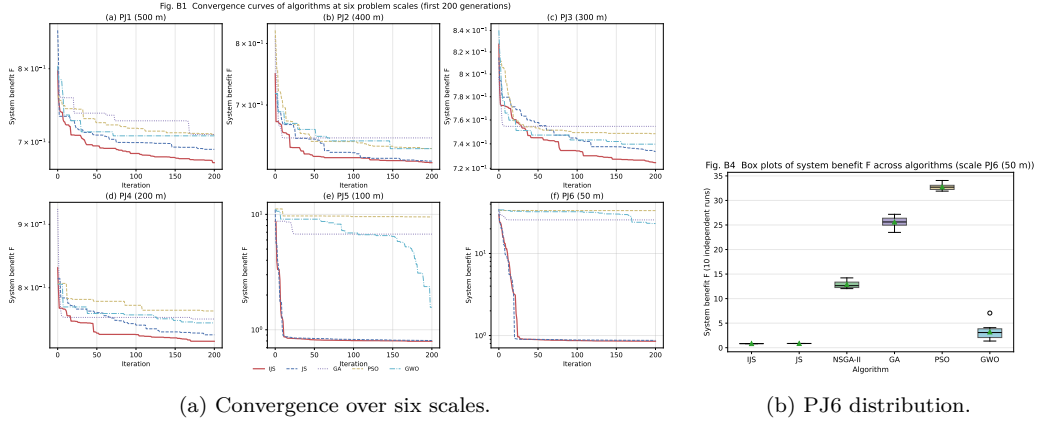


Figure 9: Historical same-generation benchmark. IJS uses roughly three population-equivalent evaluations per generation; the figure is descriptive, not an equal-budget comparison.

pattern remains relevant as a hypothesis for a current-model, equal-NFE rerun.

IJS takes about twice the wall time of one-phase competitors at PJ1–PJ5 because Lévy and DE add evaluations. The present record therefore supports only the statement that more evaluations per generation produced lower historical F . A current-model comparison must cap objective calls, pre-register paired seeds, report multiplicity-adjusted tests, effect sizes and confidence intervals, and include wall time.

5.5. Historical free-endpoint re-optimization sensitivity

All 237 points in the historical free-endpoint model converge to feasible solutions. They do not validate robustness of the current endpoint/density formulation. Within that earlier model, traffic growth dominates the energy response; cost is comparatively robust to demand, fleet and price factors

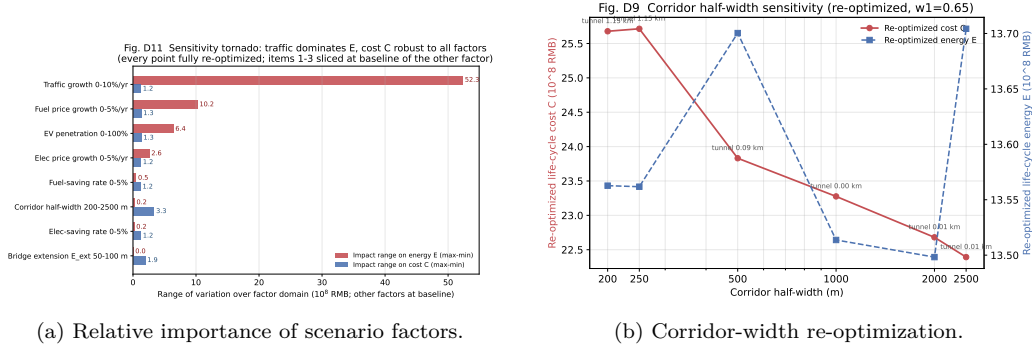


Figure 10: Historical free-endpoint sensitivity results. Every plotted point is re-optimized, but the experiment predates endpoint anchoring and density tiers.

because infrastructure cost is incurred largely at construction. Raising EV penetration moderates the high-growth energy burden, while simultaneous fuel and electricity price growth increases the monetized energy objective.

Corridor width produces a strong then saturating cost response: in the matched re-optimization experiment, C decreases from 2.568 billion RMB at 200 m to 2.239 billion RMB at 2,500 m. The benefit accelerates once the line can leave the ecological-tunnel zone, then becomes limited by urban and geometric constraints. Varying bridge approach extension E_{ext} from 50/75/100 m shifts C to 2.281/2.355/2.474 billion RMB, whereas E remains near 1.355 billion RMB. This separates an uncertain structural-cost convention from the principal energy conclusion.

6. Discussion

6.1. Engineering meaning of the improvement

The 8.33% cost decrease should not be interpreted as a generic saving for all expressways. It is a case-specific comparison under common data and prices. Its importance lies in the mechanism. M-C is slightly longer and has higher earthwork than M-A, but it reduces structural cost and energy while remaining outside Tier 2. The optimizer is therefore trading physical components, not merely exploiting route length. The endpoint constraints further prevent a candidate profile from receiving an unearned energy benefit by starting high and ending low.

The pre-density two-stage control illustrates why objective values alone are insufficient. Its lower C and E result from a line with $R_{min} = 397$ m.

347 Once feasibility is enforced, the claimed advantage disappears. Collaborative
348 search preserves the ability to relocate horizontal crossings while reshaping
349 the profile, which is particularly relevant where an OSM-triggered bridge and
350 a grade transition interact.

351 *6.2. Role of DEM and OSM*

352 DEM changes the objective landscape because the ground is sampled un-
353 der every candidate plan. Without it, lateral motion would affect only length
354 and abstract land price. OSM makes structural demand and urban exposure
355 geometry-dependent. The approach is not a substitute for survey-grade de-
356 sign: the dataset lacks bridge-deck elevations, parcel acquisition costs and
357 complete building attributes. Its value is to move early-stage optimization
358 from a one-dimensional centerline profile toward a reproducible spatial screen-
359 ing model.

360 The building tiers also clarify a policy distinction. Tier 2 is an exclusion
361 because its density exceeds what the existing expressway demonstrates as
362 traversable. Tier 1 is not forbidden; it carries a soft discouragement. This
363 avoids the unrealistic claim that an urban expressway can never pass de-
364 veloped land while still steering alternatives away from dense clusters. In
365 practice the soft term should be replaced by parcel-level acquisition and re-
366 settlement data.

367 *6.3. Algorithmic interpretation*

368 The historical ablation does not support a simplistic “chaos improves
369 everything” narrative: Tent has the smallest observed gap and DE the
370 largest. However, component activation also changes NFE, so the current
371 evidence cannot separate operator quality from budget. Tent coverage, Lévy
372 exploration and DE refinement remain plausible mechanisms to test under
373 objective-call stopping.

374 The historical benchmark suggests that feasibility should be a primary
375 algorithmic outcome. At high resolution several competitors return penalty-
376 dominated solutions, but whether IJS retains that advantage at equal NFE
377 under the current formulation is unresolved. Surrogate models, parallel eval-
378 uation or decomposition are promising routes to reduce cost after a fair base-
379 line is established.

380 *6.4. Limitations and validity threats*

381 The following limitations are material rather than cosmetic.

- 382 (1) The quasi-natural DEM is reconstructed by removing the present road in-
383 fluence, not observed before-construction terrain. Its 8.8-m pixel spacing
384 cannot resolve local drainage and retaining structures.
- 385 (2) OSM coverage is heterogeneous. Secondary roads do not trigger bridges,
386 and bridge vertical clearance is not constrained because obstacle deck
387 elevations are unavailable. A 15-degree skew limit is calibrated against
388 the existing minimum of 15.6 degrees.
- 389 (3) The Tier-1 weight (0.22 in normalized F) is a policy discouragement, not
390 a measured demolition bill. Earlier ablation, benchmark and sensitivity
391 experiments use both a pre-density and free-endpoint objective and un-
392 equal NFE; they are exploratory development records, not validation of
393 the current scheme.
- 394 (4) The energy objective is monetary and evaluates a representative travel
395 direction. Endpoint anchoring removes net-elevation arbitrage, but a
396 two-direction traffic-weighted model with calibrated regenerative braking
397 is needed for final design.
- 398 (5) Adjacent grade change is a proxy for vertical-curve smoothness. Explicit
399 crest/sag curve lengths, non-overlap, sight distance, clothoid/arc segmen-
400 tation and complete horizontal-vertical coordination checks are not yet
401 implemented.
- 402 (6) Cost rates represent early-stage estimates. Geological investigation, hy-
403 drology, parcel ownership, construction staging, embodied carbon and
404 uncertainty are outside the current model.
- 405 (7) The selected current scheme is one optimization outcome. It requires
406 multi-seed replication; algorithm claims require equal-NFE reruns, mul-
407 tiplicity control, effect sizes and confidence intervals. The stored profile
408 vector also retains one inactive legacy gene that should be removed in
409 the next experiment release.

410 These limitations define the boundary of the claim: the framework sup-
411 ports reproducible corridor-level decision and method comparison; it does
412 not replace detailed geometric or construction design.

413 7. Conclusions

414 This study presents a data-driven, collaborative horizontal-vertical high-
415 way alignment framework that connects three-dimensional geometry to life-
416 cycle cost and mixed fuel-electric traffic energy. The model extends an ear-
417 lier mathematical base with quasi-natural DEM sampling, OSM-triggered
418 bridges, an ecological tunnel zone, building-density tiers, structural substi-
419 tution and exact endpoint tie-ins. IJS supplies a corrected and empirically
420 tested search process, while feasibility and non-dominance precede entropy-
421 based decision.

422 For Guangzhou North Ring Expressway, the current $W = 500$ -m density-
423 constrained result reduces cost by 8.33% and monetized energy by 3.00%,
424 with $R_{min} = 421$ m, zero hard penalty and no Tier-2 penetration. The line is
425 slightly longer and uses more earthwork, so the improvement is attributable
426 to a transparent structural/operational trade rather than length alone. A
427 two-stage historical control shows that superficially better objectives can be
428 infeasible. Earlier ablation, benchmark and sensitivity experiments docu-
429 ment the development path, but their free-endpoint model and unequal NFE
430 prevent them from validating the current formulation. Equal-budget multi-
431 seed replication is therefore a prerequisite for confirmatory algorithm claims.

432 The next research stage should implement bidirectional energy, survey-
433 grade obstacle elevations, explicit vertical curves and sight distance,
434 parcel/embodyed-carbon objectives, uncertainty-aware costs and multi-seed
435 replication of the latest density-constrained scheme. These additions would
436 move the framework from auditable corridor planning toward design-grade
437 decision support.

438 Data and code availability

439 The manuscript is generated from the RoadPlan experiment repository.
440 GNSS and project-derived files may be subject to owner restrictions; pro-
441 cessed DEM/OSM layers, source code, fixed random seeds, result JSON
442 files and figure-generation scripts are organized for reproducibility. A
443 public archive and DOI should be inserted after anonymization review:
444 [DATA-REPOSITORY-DOI-PLACEHOLDER].

445 CRediT authorship contribution statement

446 [Author name placeholders]: Conceptualization, Methodology,
447 Software, Validation, Investigation, Data curation, Visualization, Writing—
448 original draft, Writing—review & editing. Contributions must be replaced
449 with the actual author roles before submission.

450 Declaration of competing interest

451 The authors declare that they have no known competing financial inter-
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